Kernels

- Kernels Represent a **dot product of two vectors** in a higher dimensional feature space
- Any Machine Learning Model that contains dot product of input samples can be kernelized
- Instead of calculating dot product between input vectors, calculate dot product between input vectors after mapping them to higher dimensional space

$$u \cdot v \rightarrow \text{map to } \psi(u) \text{ and } \psi(v) \rightarrow \psi(u) \cdot \psi(v) = K(u, v)$$

- Different Kernel functions are based on different mapping functions
- *Calculations of mapping and calculations of dot product in higher dimensional space are very expensive.*
- Use **Kernel Trick** to minimize both calculations (at no extra cost)